

TECHNICAL-VOCATIONAL-LIVELIHOOD (TVL) TRACK
INFORMATION & COMMUNICATION TECHNOLOGY (ICT) STRAND
COMPUTER SYSTEMS SERVICING (CSS) NC-II

The **COMPUTER SYSTEMS SERVICING NC II** Qualification consists of competencies that must possess to enable to install and configure computers systems, set-up computer networks and servers and to maintain and repair computer systems and networks.

The units of competency comprising this qualification include the following:

BASIC COMPETENCIES

Participate in workplace communication
Work in a team environment
Practice career professionalism
Practice occupational health and safety procedures

COMMON COMPETENCIES

Apply quality standards
Perform computer operations
Perform mensuration and calculation
Prepare and interpret technical drawing
Use hand tools
Terminate and connect electrical wiring and electronic circuits
Test electronic components

CORE COMPETENCIES

Install and configure computer systems
Set-up Computer Networks
Set-up Computer Servers
Maintain and Repair Computer Systems and Networks

A person who has achieved this Qualification is competent to be:

- Computer Assembler
- Computer Service Technician
- Network Technician
- Computer Maintenance Technician

TRAINEE ENTRY REQUIREMENTS

Trainees or students wishing to gain entry into this course should possess the following requirements:

- Photo ID: 2X2 (2pcs) & 1x1 (2pcs);
- Certificate of Grade 10 Completer;
- Certificate of Good Moral Character;
- Photocopy of NSO Birth Certificate;
- One brown envelope with plastic cover ;
- Can communicate either oral and written;
- Must pass the trainability/aptitude test;
- Personal Protective Equipment (apron, gloves, anti-static wrist strap, hairnet (female), mask, goggles, and sole/rubber shoes).

INSTITUTIONAL ASSESSMENT

Institutional assessment is undertaken by trainees to determine their achievement of units of competency. A certificate of achievement is issued for each unit of competency.

As a matter of policy, graduates of programs registered with DepED is not mandatory but encourage to undergo TESDA national competency assessment upon completion of the program.

TESDA NATIONAL ASSESSMENT AND CERTIFICATION ARRANGEMENTS

To attain the National Qualification of Computer Systems Servicing NC II, the candidate must demonstrate competence in all units listed in Section 1. Successful candidates shall be awarded a National Certificate signed by the TESDA Director General.

The qualification of **Computer Systems Servicing NC II** may be attained through:

Accumulation of Certificates of Competency (COCs) in all the following units of competencies:

- ✓ Install and configure computer systems
- ✓ Set-up computer networks
- ✓ Set-up computer servers
- ✓ Maintain and Repair Computer Systems and Networks

Successful candidates shall be awarded a **Certificate of Competency (COC)** in each of the core units.

Upon accumulation and submission of all COCs acquired for the relevant units of competency comprising a qualification, an individual shall be issued the corresponding **National Certificate (NC)**.

For individuals, who already possess National Certificate (NC) or Certificate of Competency (COC) along Computer Hardware Servicing NC II –

Portfolio assessment is applicable for COC #1, COC #2 and COC #4, provided the candidate is already employed and has related experience for the past three (3) years or more along the qualification. However, if the assessor finds the evidences presented inadequate, he may still require the candidate to undergo the practical demonstration or present other evidences in the form of Demonstration, Third Party Report, etc. depending on the need for supplementary evidences.

Candidates are required to take assessment for COC #3 – “Set-up computer servers”.

Assessment shall focus on the core units of competency. The basic and common units shall be integrated or assessed concurrently with the core units.

The following are qualified to apply for assessment and certification:

- Graduate of formal, non-formal, and informal including enterprise-based training programs.
- Experienced workers (wage employed or self-employed)

The guidelines on assessment and certification are discussed in detail in the Procedures Manual on Assessment and Certification and Guidelines on the Implementation of the Philippine TVET Qualification and Certification System (PTQCS).

ACKNOWLEDGEMENTS

The TRAINER wishes to extend gratitude and appreciation to Technical Education and Skills Development Authority (TESDA) -Institute of Electronics Engineers of the Philippines (IECEP), Visayas Regional Chapter, Regional Training Center (RTC), Department of Education (DepEd), and to the representatives of business, industry, academe and government agencies and labor groups who donated their time and expertise to the development of Computer Systems Servicing NC-II.

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DEFINITION OF TERMS

GENERAL

- 1) **Certification** - is the process of verifying and validating the competencies of a person through assessment
- 2) **Certificate of Competency (COC)** – is a certification issued to individuals who pass the assessment for a single unit or cluster of units of competency
- 3) **Common Competencies** - are the skills and knowledge needed by all people working in a particular industry
- 4) **Competency** - is the possession and application of knowledge, skills and attitudes to perform work activities to the standard expected in the workplace
- 5) **Competency Assessment** - is the process of collecting evidence and making judgments on whether competency has been achieved
- 6) **Competency Standard (CS)** - is the industry-determined specification of competencies required for effective work performance
- 7) **Context of Assessment** - refers to the place where assessment is to be conducted or carried out
- 8) **Core Competencies** - are the specific skills and knowledge needed in a particular area of work - industry sector/occupation/job role
- 9) **Critical aspects of competency** - refers to the evidence that is essential for successful performance of the unit of competency
- 10) **Elective Competencies** - are the additional skills and knowledge required by the individual or enterprise for work
- 11) **Elements** - are the building blocks of a unit of competency. They describe in outcome terms the functions that a person performs in the workplace.
- 12) **Evidence Guide** - is a component of the unit of competency that defines or identifies the evidences required to determine the competence of the individual. It provides information on critical aspects of competency, underpinning knowledge, underpinning skills, resource implications, assessment method and context of assessment
- 13) **Level** - refers to the category of skills and knowledge required to do a job
- 14) **Method of Assessment** - refers to the ways of collecting evidence and when, evidence should be collected
- 15) **National Certificate (NC)** – is a certification issued to individuals who achieve all the required units of competency for a national qualification defined under the Training Regulations. NCs are aligned to specific levels within the PTQF
- 16) **Performance Criteria** - are evaluative statements that specify what is to be assessed and the required level of performance
- 17) **Qualification** - is a cluster of units of competencies that meets job roles and is significant in the workplace. It is also a certification awarded to a person on successful completion of a course in recognition of having demonstrated competencies in an industry sector
- 18) **Range of Variables** - describes the circumstances or context in which the work is to be performed

- 19) **Recognition of Prior Learning (RPL)** – is the acknowledgement of an individual's skills, knowledge and attitudes gained from life and work experiences outside registered training programs
- 20) **Resource Implications** - refers to the resources needed for the successful performance of the work activity described in the unit of competency. It includes work environment and conditions, materials, tools and equipment
- 21) **Basic Competencies** - are the skills and knowledge that everyone needs for work
- 22) **Training Regulations (TR)** – refers to the document promulgated and issued by TESDA consisting of competency standards, national qualifications and training guidelines for specific sectors/occupations. The TR serves as basis for establishment of qualification and certification under the PTQF. It also serves as guide for development of competency-based curricula and instructional materials including registration of TVET programs offered by TVET providers
- 23) **Underpinning Knowledge** - refers to the competency that involves in applying knowledge to perform work activities. It includes specific knowledge that is essential to the performance of the competency
- 24) **Underpinning Skills** - refers to the list of the skills needed to achieve the elements and performance criteria in the unit of competency. It includes generic and industry specific skills
- 25) **Unit of Competency** – is a component of the competency standards stating a specific key function or role in a particular job or occupation; it is the smallest component of achievement that can be assessed and certified under the PTQF

SECTOR SPECIFIC

1. **Computer System** - The complete computer made up of the CPU, memory and related electronics (main cabinet), all the peripheral devices connected to it and its operating system. Computer systems fall into two broad divisions: clients and servers. Client machines fall into three categories from low to high end: laptop, desktop and workstation. Servers range from small to large: low-end servers, midrange servers and mainframes.
2. **Computer Network**, or simply a **Network**, is a collection of computers and other hardware interconnected by communication channels that allow sharing of resources and information. Where at least one process in one device is able to send/receive data to/from at least one process residing in a remote device, then the two devices are said to be in a network. It is a group of devices connected to each other. Networks may be classified into a wide variety of characteristics, such as the medium used to transport the data, communications protocol used, scale, topology, benefit, and organizational scope.
3. **Configuration** - The makeup of a system. To "configure" is to choose options in order to create a custom system. "Configurability" is a system's ability to be changed or customized.
4. **Connector** - Any plug and socket that links two devices together. Although taken for granted and rarely in the limelight, connectors are a huge industry, and the quality of these components is more critical than most people would imagine. When not designed or constructed properly, they often become the weakest element in an electronic system.
5. **Display Adapter** - A plug-in card in a desktop computer that converts the images created in the computer to the electronic signals required by the monitor. It

determines the maximum resolution, refresh rate and number of colors that can be displayed, which the monitor must also be able to support. On many PC motherboards, the display adapter circuits are built into the chipset, and an AGP or PCI card is not required.

6. **Expansion Board** - A printed circuit board that plugs into an expansion slot and extends the computer's capability to control a peripheral device. All the boards (cards) that plug into a computer's bus are expansion boards, such as display adapters, disk controllers, network adapters and sound cards.
7. **Expansion Bus** - An input/output bus typically comprised of a series of slots on the motherboard. Expansion boards (cards) are plugged into the bus. ISA and PCI are the common expansion buses in a PC.
8. **Graphical User Interface** - A graphics-based user interface that incorporates movable windows, icons and a mouse. The ability to resize application windows and change style and size of fonts are the significant advantages of a GUI vs. a character-based interface. GUIs have become the standard way users interact with a computer, and the major GUIs are the Windows and Mac interfaces along with Motif for Unix and the GNOME and KDE interfaces for Linux.
9. **LAN** - a **local area network (LAN)** is a computer network that interconnects computers in a limited area such as a home, school, computer laboratory, or office building using network media. The defining characteristics of LANs, in contrast to wide area networks (WANs), include their usually higher data-transfer rates, smaller geographic area, and lack of a need for leased telecommunication lines.
10. **Motherboard** - Also called the "system board," it is the main printed circuit board in an electronic device, which contains sockets that accept additional boards. In a desktop computer, the motherboard contains the CPU, chipset, PCI bus slots, AGP slot, memory sockets and controller circuits for the keyboard, mouse, disks and printer. It may also have built-in controllers for modem, sound, display and network, obviating the need to plug in a card.
11. **Networks** – see Computer Network.
12. **Operating System** - The master control program that runs the computer. The first program loaded when the computer is turned on, its main part, the "kernel," resides in memory at all times. The operating system sets the standards for all application programs that run in the computer. The applications "talk to" the operating system for all user interfaces and file management operations.
13. **Peripheral** -Any hardware device connected to a computer, such as a monitor, keyboard, printer, disk, tape, graphics tablet, scanner, joy stick, paddle or mouse
14. **Server** - A computer system in a network that is shared by multiple users. Servers come in all sizes from x86-based PCs to IBM mainframes. A server may have a keyboard, monitor and mouse directly attached, or one keyboard, monitor and mouse may connect to any number of servers via a KVM switch. Servers may be also be accessed only through a network connection as well.
15. **Sound Card** - Also called a "sound board" or "audio adapter," it is a computer expansion board that records and plays back sound, providing inputs from a microphone or other sound source and outputs to speakers or an external amplifier. The de facto standard for sound card compatibility in PCs is Creative Labs' Sound Blaster.
16. **User Interface** - All graphics based today, the user interface includes the windows, menus and method of interaction between you and the computer. Prior to the Mac, Windows and Motif (UNIX) interfaces, all interaction was based on commands entered by the user. Operating systems may support optional interfaces and allow a new shell, or skin, to be used instead.

17. **Virus** - Software used to infect a computer. After the virus code is written, it is buried within an existing program. Once that program is executed, the virus code is activated and attaches copies of itself to other programs in the system. Infected programs copy the virus to other programs.
18. **WAN - a Wide Area Network** (WAN) is a network that covers a broad area (i.e., any telecommunications network that links across metropolitan, regional, or national boundaries) using private or public network transports. Business and government entities utilize WANs to relay data among employees, clients, buyers, and suppliers from various geographical locations. In essence, this mode of telecommunication allows a business to effectively carry out its daily function regardless of location
19. **WEEE Directives** - the prevention of waste electrical and electronic equipment (WEEE), and in addition, the reuse, recycling and other forms of recovery of such wastes so as to reduce the disposal of waste. It also seeks to improve the environmental performance of all operators involved in the life cycle of electrical and electronic equipment, e.g. producers, distributors and consumers and in particular those operators directly involved in the treatment of waste electrical and electronic equipment.in memory at all times. The operating system sets the standards for all application programs that run in the computer. The applications "talk to" the operating system for all user interfaces and file management operations.
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What is a Computer ?

- a programmable electronic device that executes the instructions in a program
- A Computer has four Functions:
 - a. Accepts Data
 - b. Process Data
 - c. Produces Output
 - d. Stores Results



Classes of computers

- Micro computers
- Minicomputers or Midrange Computers
- Mainframe Computers
- Servers
- Workstation
- Portable Data Entry Terminals
- Embedded computer systems
- Super Computer

Micro Computers

- are the most common type of computers in existence today, it was introduced with the advent of single chip large scale integrated circuit computer processors
- These computers include Desktop computers, Personal digital assistants (more common known as PDA's), Palmtop computers, Laptop and notebook computers



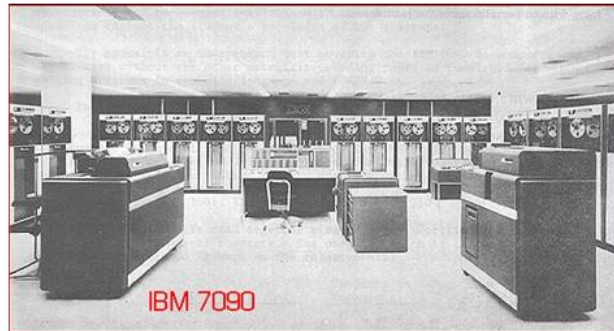
Minicomputers or Midrange Computers

- The term "Mini computer" was coined at the time when most computers were cabinet sized
- Mini computers were much smaller, less powerful, and much less expensive
- The first Mini computers generally only performed one task at a time, while bigger computers ran multi-tasking operating systems, and served multiple users



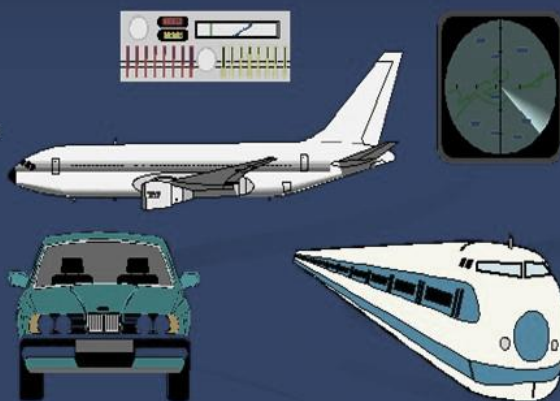
Mainframe Computers

- also known as "big iron" which refers to large, expensive, ultra-fast computers
- are computers used mainly by large organizations for critical applications, typically bulk data processing such as census, industry and consumer statistics, ERP (Enterprise Resource Planning), and financial transaction processing.
- A very large and expensive computer capable of supporting hundreds, or even thousands, of users simultaneously



Embedded computer systems

- Embedded computers are general purpose CPUs that are a part of a machine or device.
- This computer runs a program that is stored in read only memory (ROM) and is only intended to operate a specific machine or device.
- Embedded systems are typically required to operate continuously without being reset or rebooted, and once employed in their task the software usually cannot be modified
- An automobile, washing machine and a DVD player



Super Computer

- A supercomputer is focused on performing one task involving intense numerical calculations such as weather forecasting and solving scientific problems.
- It has processing speeds of up to billions and billions gigabytes per second.
- Most supercomputers run on a Linux or Unix operating system, as these operating systems are extremely flexible, stable, and efficient.



Lightning speed

Supercomputers, the fastest type of computer, are very expensive and used for specialized applications that require large amounts of calculations.

How many?
Calculations per second:
Gigaflop 1 billion
Teraflop 1 trillion
Petaflop 1 quadrillion
Exaflop 1 quintillion

Year	Performance	Maker
1993	16 gigaflops 16,000,000,000	Thinking Machine C-5
1994	236 gigaflops 236,000,000,000	Fujitsu Numerical Wind Tunnel
1996	368 gigaflops 368,000,000,000	Hiatchi CP-PACS
1997	1.8 teraflops 1,800,000,000,000	Intel Asci Red
2000	12.3 teraflops 12,300,000,000,000	IBM Asci White
2002	36 teraflops 36,000,000,000,000	NEC Earth Simulator
2004	280 teraflops 280,000,000,000,000	IBM Blue Gene/L
2008	1 petaflop 1,000,000,000,000,000	IBM Blue Gene/P
2011	10 petaflops 10,000,000,000,000,000	IBM Blue Gene/P
2017	20 petaflops 20,000,000,000,000,000	IBM Blue Gene/P
2022*	1 exaflop 1,000,000,000,000,000,000	To be determined

*In development. IBM has not set a timeline
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TYPES OF INPUT DEVICES

KEYBOARD

- is an input device designed to enter text, characters and other commands into the computer.

AT KEYBOARD



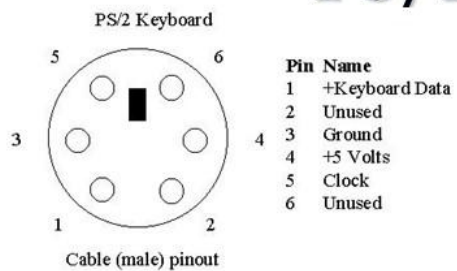
AT keyboard connector (DIN5)



Connector Pin #	Purpose
Pin 1	KBDCLK (clock)
Pin 2	KBDAT (data)
Pin 3	KBRST (reset, not used)
Pin 4	GND
Pin 5	VCC (+5V)



PS/2 KEYBOARD



Personal System/2 or **PS/2** is used for connecting some keyboards and mice to a PC compatible computer system.



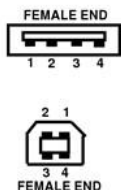
DIN (Deutsche Industrial Normale) - is a series of uniformity standards developed in Germany, which apply to commonly manufactured items.



USB KEYBOARD



PC USB Connectors Pinout
1 = +5 Volts
2 = -Data
3 = +Data
4 = Ground



Type	Port Image	Connector Image
Type A	4.5mm x 12.0mm 	
Type B	7.3mm x 8.5mm 	
Mini-A	3.0mm x 6.8mm 	
Mini-B	3.0mm x 6.8mm 	

INFRARED / IR KEYBOARD



- Infrared Data Association (IrDA)
- defines physical specifications communications protocol standards for the short-range exchange of data over infrared light
- The "IR" in its name indicates the communication method this keyboard uses; it's infra-red. That means it needs line of sight between the two transmitter LEDs under the dark plastic cover on the keyboard and its receiver unit - but, like many remote controls, it's not **very** touchy about alignment of the two components, as long as they're not too far apart.



Bluetooth™ KEYBOARD

- was developed in 1994 by Jaap Haartsen and Sven Mattisson, who were working for Ericsson Mobile Platforms
- Bluetooth was named after a tenth-century king, Harald Bluetooth, King of Denmark and Norway
- The Bluetooth logo merges the Germanic runes analogous to the modern Latin letter H and B: (for Harald Bluetooth) * (Hagall) and B (Berkanan) merged together, forming a bind rune.
- is a wireless protocol utilizing short-range communications technology facilitating data transmission over short distances from fixed and/or mobile devices



CONVERTERS



AT to PS/2 Connector



PS/2 to AT Connector



PS/2 to USB Connector



USB to PS/2 Connector



MOUSE

SERIAL MOUSE



Serial Port



How to clean your trackball mouse

- Turn the mouse upside down.
- Rotate the retaining ring on the bottom of the mouse counter-clockwise, then remove the retaining ring and mouse ball.
- Remove any dust, lint, or dirt from the mouse ball with a soft cloth.
- Clean the mouse rollers with a cotton swab dipped in isopropyl alcohol.
- Replace the mouse ball and lock the retaining ring into place



PS2 MOUSE



USB MOUSE



IR MOUSE



BLUETOOTH MOUSE



CONVERTERS



USB to Serial Converter



PS/2 to Serial Converter



Serial to PS/2 Converter

IMAGE SCANNERS



DIGITAL CAMERAS



TRACKBALL



PEN OR STYLUS



JOYSTICK



STEERING WHEEL



GAMEPAD



TOUCH SCREEN



TOUCH PAD



BAR CODE READER



CARD READER



TYPES OF OUTPUT DEVICES

MONITOR

- is a computer peripheral device which is capable of showing video output to the user
- displays alphanumeric text, symbols such as icons, and graphics such as images

Guidelines for suitable resolutions for different monitors:

- 14 inch monitor is adequate for 800 x 600 resolution.
- 15 inch monitor is adequate for 1024 x 768 resolution.
- 17 inch monitor is adequate for 1024 x 768 resolution.
- 19 inch monitor is adequate for 1280 x 1024 resolution.
- 21 inch monitor is adequate for 1600 x 1280 resolution.

MONITOR



CRT (*Cathode ray tube*)



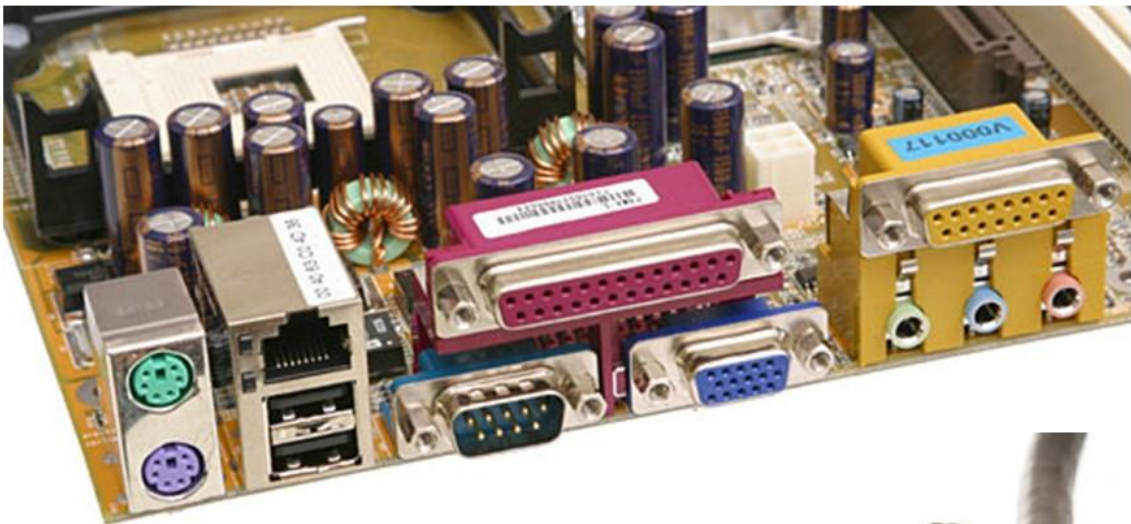
CRT (*Cathode ray tube*) Flat



LCD (Liquid crystal display)

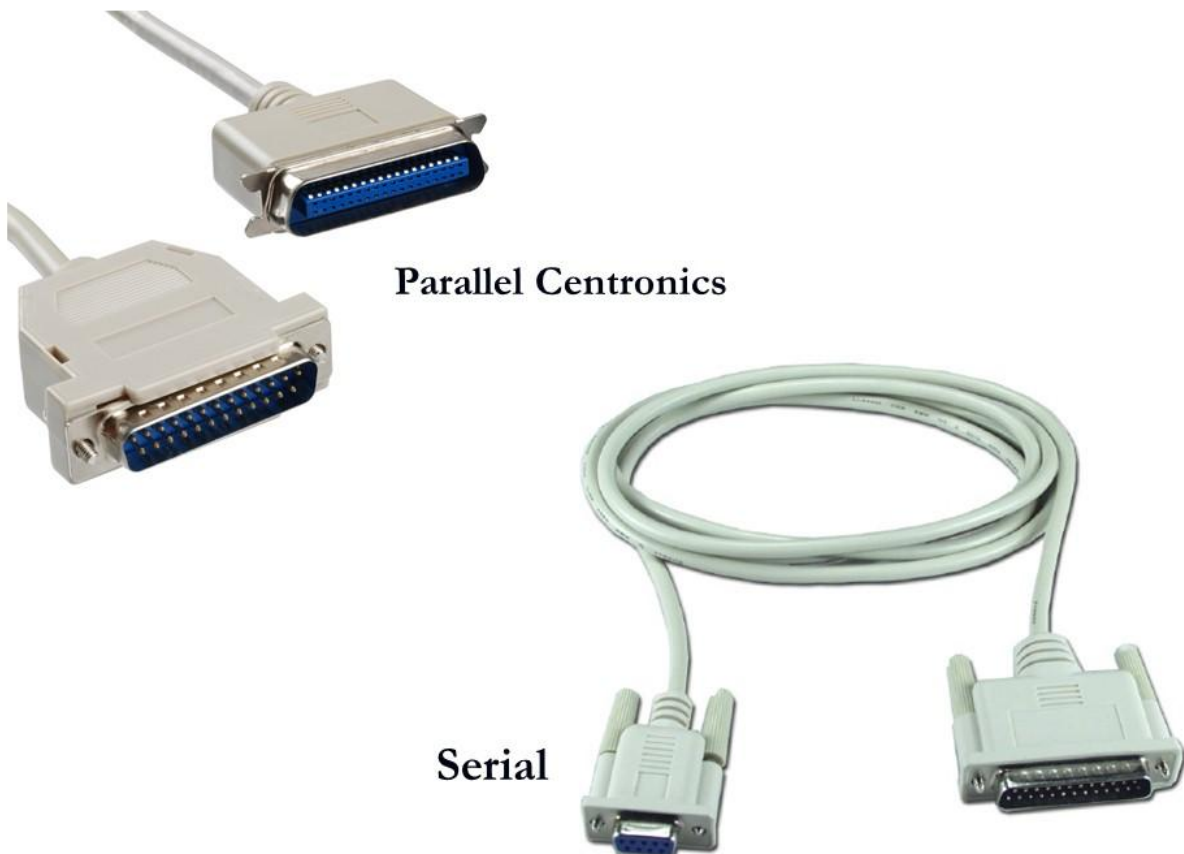


LCD (Liquid crystal display) Wide Screen



Printer

- A device that prints text or illustrations on paper.
- it vary in size, speed, sophistication, and cost
- most common I/O interface for printers has been the parallel Centronics interface with a 36-PIN plug. In the future, however, new printers and computers are likely to use a serial interface, especially Universal Serial Bus or FireWire





USB

Firewire



4-pin connector



6-pin connector

DOT-MATRIX PRINTER

- A type of **printer** that produces **characters** and illustrations by striking **pins** against an ink ribbon to print closely spaced **dots** in the appropriate shape
- Each pin makes a **dot**, and combinations of dots form characters and illustrations
- vary from about 50 to over 500 cps



INK-JET PRINTER

- is a computer peripheral that produces hard copy by spraying ink onto paper.
- typical ink-jet printer provides a resolution of 300 dpi (dots per inch), although some newer models offer higher resolutions.
- Low-end inkjets use three ink colors (cyan, magenta and yellow)
- Normally inkjet and laser printers have 3 levels of quality settings: draft, normal, best
- typical inkjet printer may vary between 1 to 28 ppm for black text and 1 to 20 ppm for color photo or graphics.



LASER PRINTER

- is a common type of computer printer that rapidly produces high quality text and graphics on plain paper.
- The fastest color laser printers can print over 100 pages per minute (6000 pages per hour).



PLOTTER

- is a vector graphics (also called **geometric modeling or object-oriented graphics**) printing device that connects to a computer
- they draw lines using a pen. As a result, they can produce continuous lines, whereas printers can only simulate lines by printing a closely spaced series of dots.
- are considerably more expensive than printers. They are used in engineering applications where precision is mandatory



SPEAKER



LCD PROJECTOR



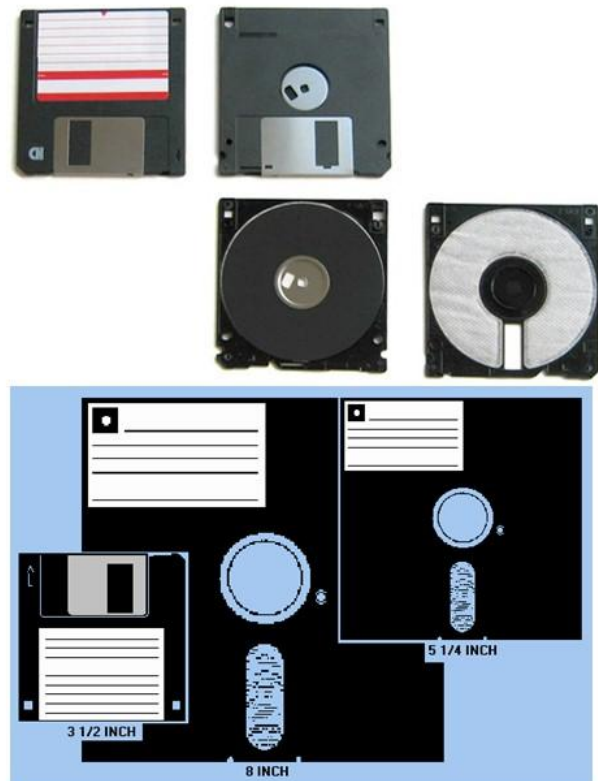
HEADPHONES



TYPES OF STORAGE DEVICES

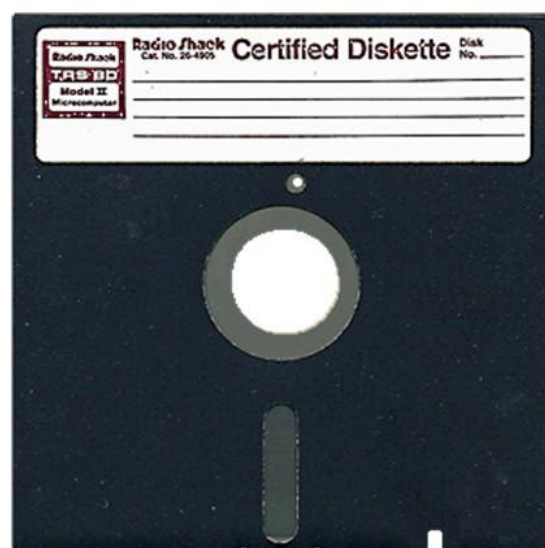
FLOPPY DISK

- A reusable magnetic storage medium introduced by IBM in 1971
- It is called *floppy* because it flops if you wave it
- Disk drives for floppy disks are called *floppy drives*



8 INCH DISKETTE

- In 1971, IBM introduced the 8-inch floppy disk, initial capacity was about 100K bytes
- In 1979 the Radio Shack **TRS-80 II** computer system had an internal 8-inch floppy drive capable of storing 500K of data.



5 1/4 INCH DISKETTE

- In 1976, Shugart introduced the 5 1/4-inch floppy disk. Initial capacity was about 100K, eventually reaching 1.2M bytes per disk

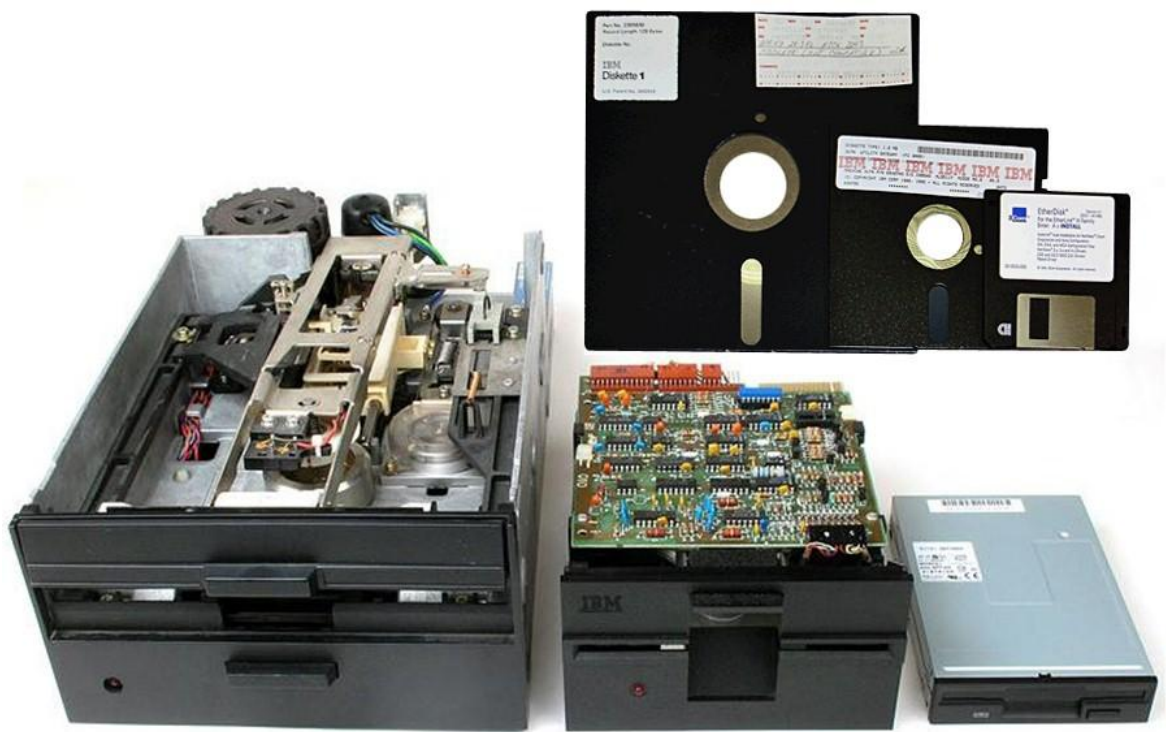


3 1/2 INCH DISKETTE

- In 1980, Sony introduced the 3 1/2-inch floppy disk. Initially holding about 400K, current capacity is 1.4Meg per disk
- 720K double density
- 1.44MB high density



FLOPPY DISK DRIVE



External Floppy Disk Drive



Zip drive

- introduced by Iomega in late 1994
- is a medium-capacity removable disk storage system
- Originally it had a capacity of 100 MB, but later versions increased this to first 250 MB and then 750 MB
- Zip drives are available in multiple interfaces including usb 1.1, paralell port, (Small Computer System Interface) SCSI, ATA, and parallel port.



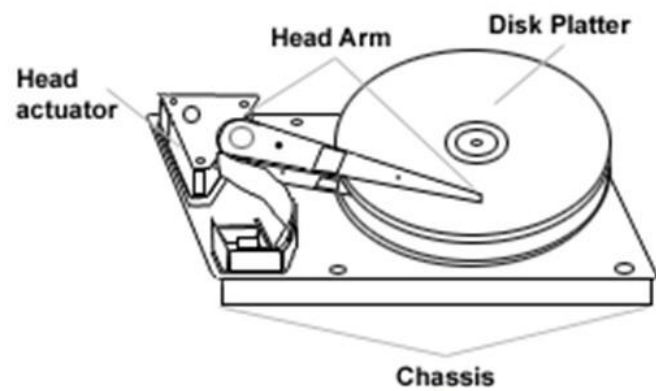
HARD DISK

- A hard disk drive (HDD), commonly referred to as a hard drive, hard disk or fixed disk drive
- A magnetic disk on which you can store computer data
- Mass Storage is measured in kilobytes, megabytes, gigabytes and terabytes



The four main components of a hard disk

- **Platter** - The actual fixed disk within the hard disk drive
- **Head actuator** - controls the **head arm** which reads the information off of the disk platter
- And the **chassis** encases and holds all the hard disk drive components



TYPES Of Hard Disk Interface

- **ESDI** - Enhanced Small Disk Interface
- **SCSI** - Small Computer System Interface,
- **IDE** - Integrated Drive Electronics
- **SATA** - Serial Advanced Technology Attachment

ESDI

Enhanced Small Disk Interface



- was a disc interface designed by Maxtor Corporation in the early 1980s
- 34-pin common control cable, and a 20-pin data channel cable for each device

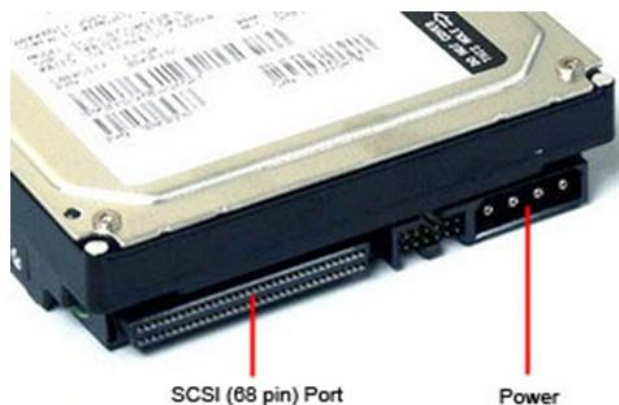


SCSI

Small Computer System Interface



- SCSI was derived from "SASI", the "Shugart Associates System Interface", introduced by that company in 1981

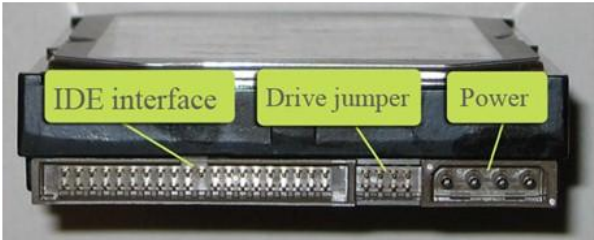


SCSI Controller



IDE Integrated Drive Electronics

- **Master or single-drive:** The drive is shipped configured for a master or a single-drive with a jumper set on pins 7 and 8.
- **Drive is slave:** To configure the drive as a slave, or second drive on the cable, remove all the jumpers.
- **Master with non-ATA compatible slave:** Use this setting if the slave drive is not recognized. Configure the master drive with a jumper set on pins 5 and 6 and pins 7 and 8 to enable this option.
- **Alternate capacity jumper:** Use this jumper only if your system cannot recognize drives with capacities greater than 2.1 Gbytes.



Options jumper block

	Master or single drive
	Drive is slave
	Master with non ATA-compatible slave
	Cable select
	Limit drive capacity 40 Gbytes = 32 GB <40 Gbytes = 2.1 Gbytes

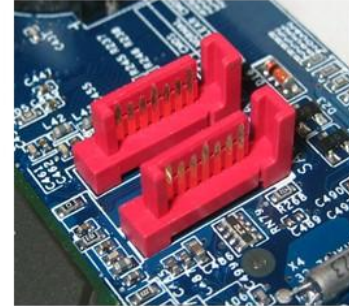
7 5 3 1
8 6 4 2

Circuit Board



SATA

Serial Advanced Technology Attachment



OPTICAL DISK

- Originally developed in the late 1960s
- is a random access storage medium
- A storage medium from which data is read and to which it is written by lasers



CD-R and CD-RW



12 CM



12 CM



8 CM



8 CM

What capacities of blank CD-R and CD-RW discs are available?

- Manufacturers commonly express disc capacity in terms of how much **Red Book digital audio** (in minutes) and **computer data** (in megabytes) a disc can contain
- **Red Book** is the standard for audio CDs (Compact Disc Digital Audio system, or CDDA). It is named after one of a set of color-bound books that contain the technical specifications for all CD and CD-ROM formats.
- Historically, **63 minute/550 MB (12 cm)** and **18 minute/158 MB (8 cm)** discs were once available but are now rendered obsolete by advances in recording technology. Currently, **74 minute/650 MB**, **80 minute/700 MB (12 cm)** and **21 minute/185 MB (8 cm)** discs are the market standards.



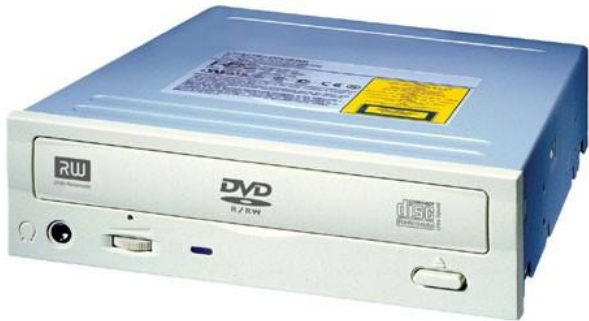
CD ROM

- Compact Disc Read-Only Memory
- was originally designed for music storage and playback
- May be connected to the computer via an IDE (ATA), SCSI, S-ATA, Firewire, or USB interface
- Transfer rate : 63.8976 Mb/s



DVD

- **Digital Video Disc**, and later **Digital Versatile Disc**
- Capacities for single sided is 4.7GB's for single layer and 8.5GB's for dual-layer disks
- Capacities for double sided is 9.4GB's for single layer and 17GB's for dual-layer disks
- Transfer rate: **11.08Mbps**



Four DVD disc construction formats:

- 1. Single-sided, single-layered - Also known as **DVD-5**, this simplest construction format holds **4.7 Gigabytes** (GBytes) of digital data. The "5" in "DVD-5" signifies the nearly 5 GBytes worth of data capacity
- 2. Single-sided, dual-layered - The **DVD-9** construction holds about **8.5 GBytes**. DVD-9s do *not* require manual flipping: the DVD player automatically switches to the second layer in a fraction of a second, by re-focusing the laser pickup on the deeper second layer
- 3. Double-sided, single-layered - Known as **DVD-10**, this construction features a capacity of **9.4 GBytes** of data. Almost all DVD players require you to manually flip the DVD, that's why the DVD-10 is called the "flipper" disc.
- 4. Double-sided, dual-layered - The **DVD-18** construction can hold approximately **17 GBytes** or about 8 hours of video and audio as a DVD-Video. To access the content on the other side of a DVD-18, you have to manually flip the DVD

DVD-R

- **DVD-R is (pronounced "dash R" not "minus R")**
- DVD-R format was developed by Pioneer and was released in the second half of 1997
- Companies that support DVD-R include Pioneer, **Toshiba, Hitachi, and Panasonic**
- is a non-rewriteable format and it is compatible with about **93%** of all DVD Players and most DVD-ROMs.

DVD+R

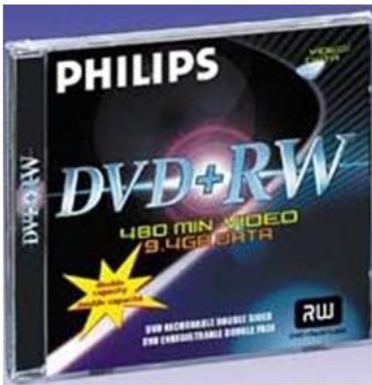
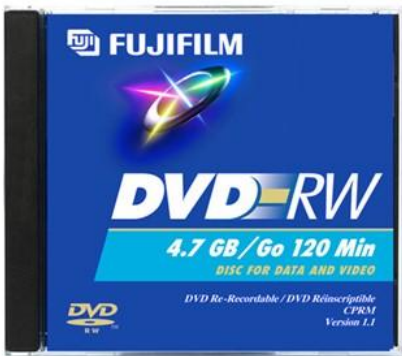
- **DVD+R** (pronounced "DVD plus R")
- DVD+R was developed by Sony and Philips and was introduced in 2002
- DVD+R include Sony, Philips, **Hewlett-Packard, Ricoh, and Yamaha**
- is a non-rewritable format and it is compatible with about **89%** of all DVD Players and most DVD-ROMs

What is the difference between the DVD-R and DVD+R formats?

- The only difference between the formats is the way they determine the location of the laser beam on the disc. **DVD-R** discs use tiny marks along the grooves in the discs, called **land prepits**, to determine the laser position. **DVD+R** discs do not have **land prepits**, but instead measure the "wobble frequency" as the laser moves toward the outside of the disc.

DVD+RW / DVD-RW

- Short for **DVD-ReWritable**
- a re-recordable **DVD** format which can be erased and recorded over numerous times without damaging the medium



DVD+R DL and DVD-R DL Double Layer (DL)

- Also called Dual Layer writeable
- These discs are only writable on one side of the disc, but contain two layers on that single side for writing data to
- They can hold up to 8.5GB on the two layers



Figure 1 – Cross Section of a DVD+R DL Double Layer Disc



Double-Sided DVD

- also come in two formats: DVD-R and DVD+R, including the rewritable DVD-RW and DVD+RW
- Double-Sided discs include a single layer on each side of the disc that data can be recorded to. These discs can hold about 8.75GB of data if you burn to both sides



DVD-RAM

- DVD – Random Access Memory
- is a disc specification presented in 1996 by the DVD Forum
- is optical storage that can be re-written hundreds of thousands of times and has an expected media life of 30 years
- offers capacity of 4.7GB per side/9.4GB per double-sided and is available in both single-sided and double-sided media



HD DVD

- **High-Definition Digital Versatile Disc**
- is a high-density optical disc format designed for the storage of data and high-definition video
- was designed principally by Toshiba, and was to be the successor to the standard DVD format
- However, in February, 2008, Toshiba abandoned the format, announcing it would no longer develop or manufacture HD DVD players
- As of February 2008, Toshiba has announced plans to discontinue development, marketing and manufacturing while still providing product support and after-sale service to consumers of the format. Toshiba will continue support by holding spare parts for eight years after sales have stopped
- **Transfer rate : 36Mbps**





- Holographic Versatile Disc
- It employs a technique known as collinear holography, whereby two lasers, one red and one green, are collimated in a single beam.
- is an optical disc technology that would hold up to 3.9 terabytes (TB) of information.
- The HVD also has a transfer rate of 1 Gbit/s (125 MB/s).



USB FLASH DRIVE

- introduced in early 2000
- A small, portable flash memory card that plugs into a computer's USB port and functions as a portable hard drive
- are also called *thumb drives*, *jump drives*, *pen drives*, *key drives*, *tokens*, or simply *USB drives*
- connected by USB 1.1 or USB 2.0 or both



Memory Cards

- is a solid-state electronic flash memory data storage device used with digital cameras, handheld computers also known as palmtop computers and Mobile computers, telephones, music players, video game consoles, and other electronics.
- **SD - Secure Digital**, is a non-volatile memory card format developed by Matsushita, SanDisk, and Toshiba for use in portable devices.
- **MMC - Multimedia Card** is a flash memory card standard.



Memory Card Reader

